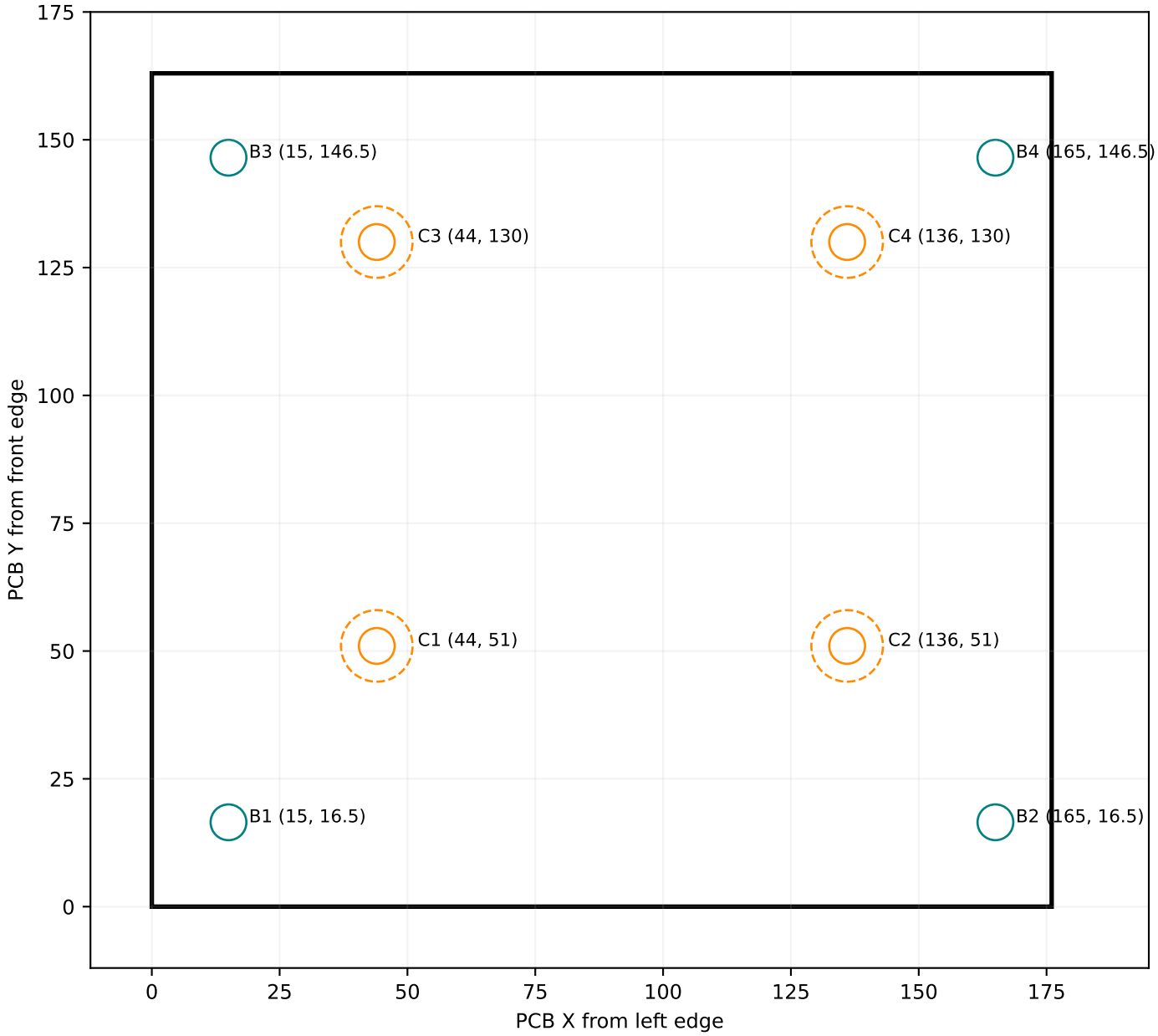
01 — Datum, board supports and cooler tie-down pattern



PCB 176 × 163
B1–B4: carrier-to-housing
150 × 130 pitch, M3 inserts

C1–C4: cooler tie-down
92 × 79 pitch
D14 printed column (dashed)
Screw from below the floor,
through PCB, into the cooler.

Enclosure translation:
PCB underside → (15, 15, 20)
P9 template has both patterns.

02 — Vertical stack

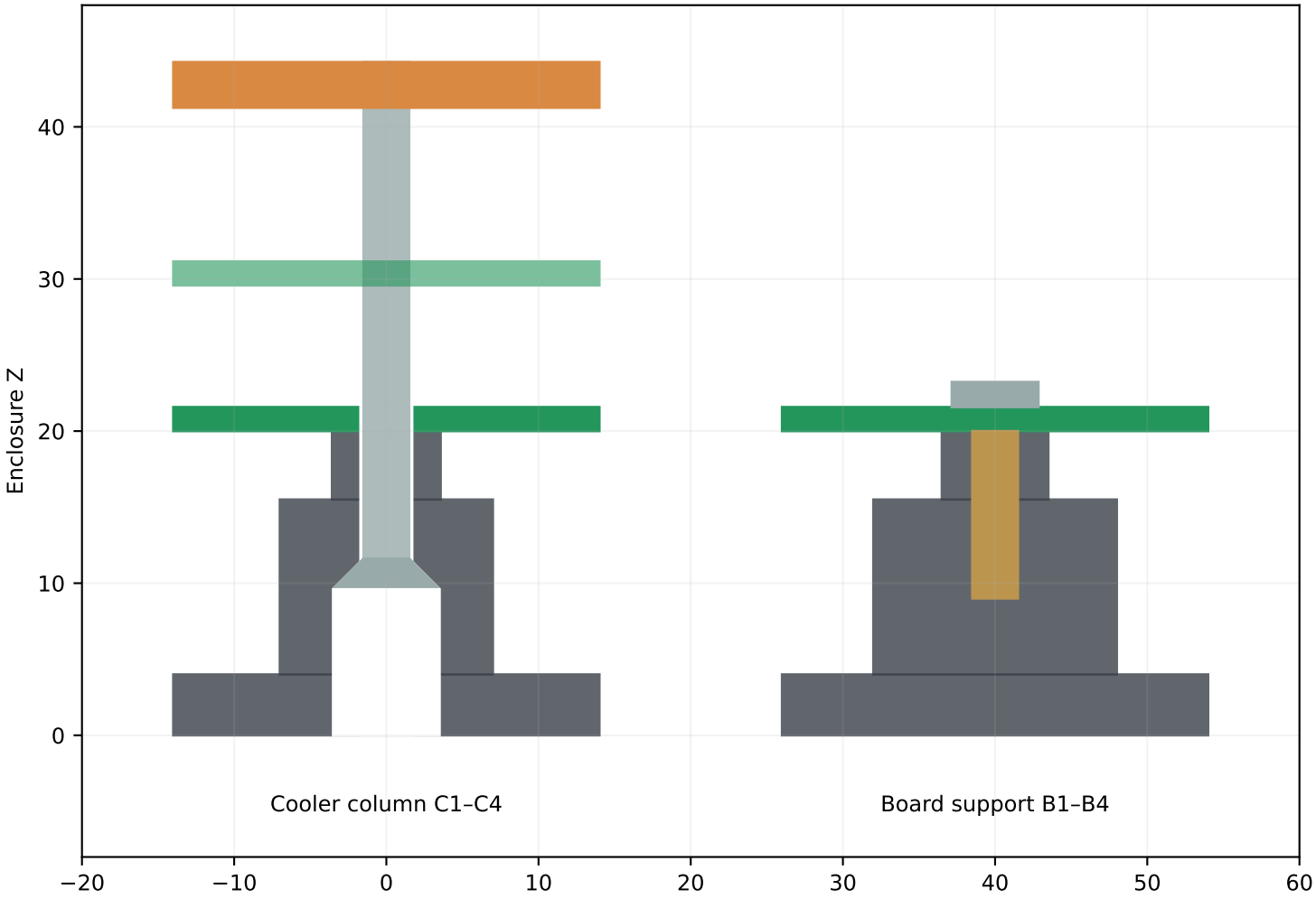


PCB underside 20.000
PCB top 21.578
Module underside 29.578
TTP/cooler seat 41.258
Reference cooler top 91.558
Roof underside 93.500
Cooler → roof 1.942

Underside clearance 9.5
(lowest part −6.5 below PCB)

GMSL PCB top 39.178 is 1.58 below the cooler seat where they overlap by 4.5 mm in plan: measure before printing P3.

03 — Cooler tie-down column and board support



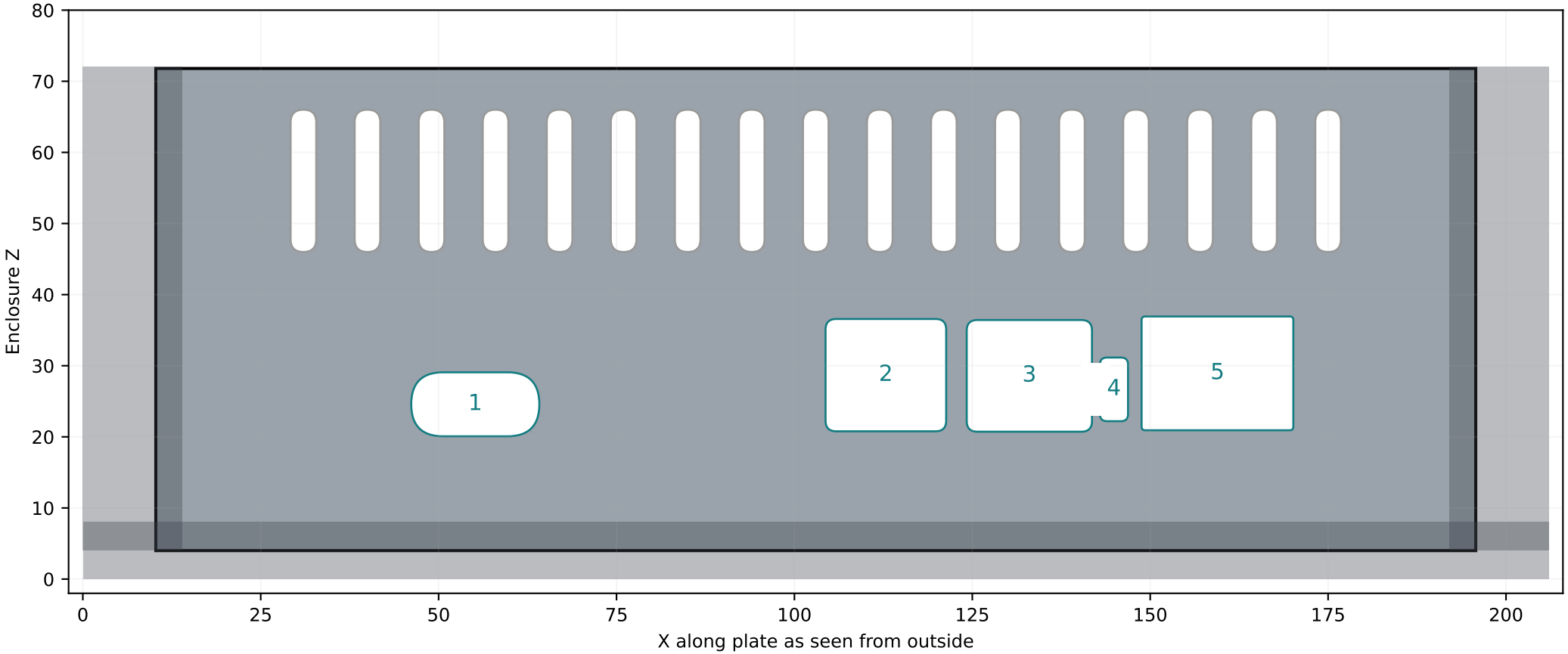
Cooler tie-down (user request)
D14 column, D7.14 landing
D3.4 clearance hole
D7 pocket, 90° countersink
seat raised to Z9.76 in the column

Load path: floor → column → carrier
→ module → heatsink standoffs

SCREW: M3 × 35 flat head. AGX top
Z40.76 + 4 mm (8 threads) − seat
Z9.76 = 35.0. Thermal pad
0.5 mm lifts the heatsink only.

Board support: D16 / D7 land,
M3 × 10 insert, M3 × 10 button.

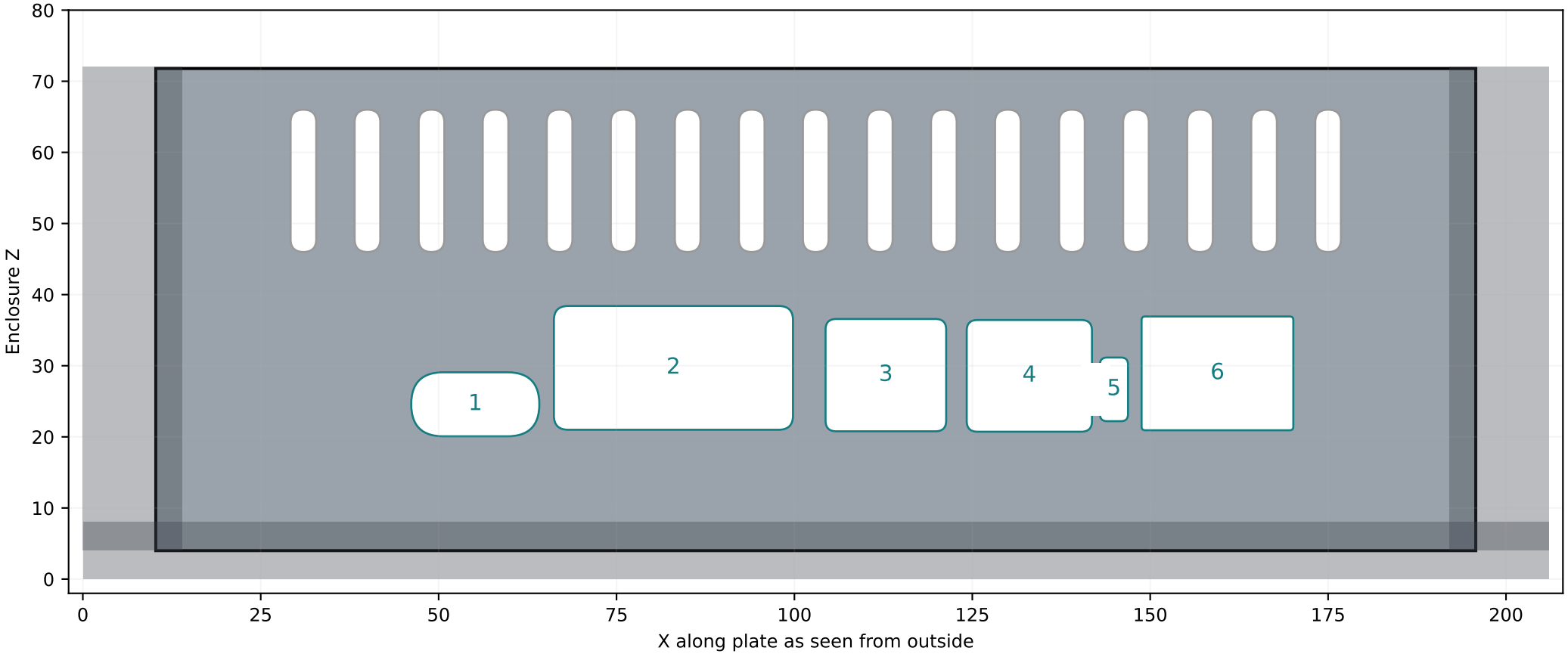
04 — Front plate WITHOUT GMSL (viewed from outside)



1. audio: enclosure X 46.15..64.15, Z 20.08..29.08, r4.5. One obround over both jacks, centred on the jack axes; sized for two D8 plug overmolds + 0.5.
2. lan_1g: enclosure X 104.38..121.33, Z 20.78..36.57, r1.5. Centred on the J5 face; clears the RJ45 plug body and latch lever on insertion + 0.5; face LEDs visible.
3. lan_10g: enclosure X 124.22..141.82, Z 20.73..36.43, r1.5. Centred on the J12 face; clears the RJ45 plug body and latch lever on insertion + 0.5.
4. led: enclosure X 142.94..146.88, Z 22.20..31.15, r1.0. LED4 light pipes x 128.437..131.382, z 2.703..10.653 at y -1.5 (+0.5); bridged to neighbours when the web is < min_panel_web. Measured by slicing references/hardware/carrier-full.stl (+ open shells) at the mating face, 2026-10-25; pins and pegs below the PCB excluded.
5. dc_in: enclosure X 148.79..170.11, Z 20.93..36.93, r0.5. Sized and centred on the mated screw-flange plug, which is 15 mm tall on an 8.4 mm header and passes through the plate (body 9.9 mm outside the header face). Wire the plug before inserting it: its clamp screws face up.

3 mm plate printed exterior face down, 0.6 × 45° lead-in on every window. Plate inner face 0.37 (front) / 0.25 (rear) from the most protruding carrier part.
Shaded: corner posts and sill that frame the plate. Windows narrower than a 1.8 mm web are bridged.

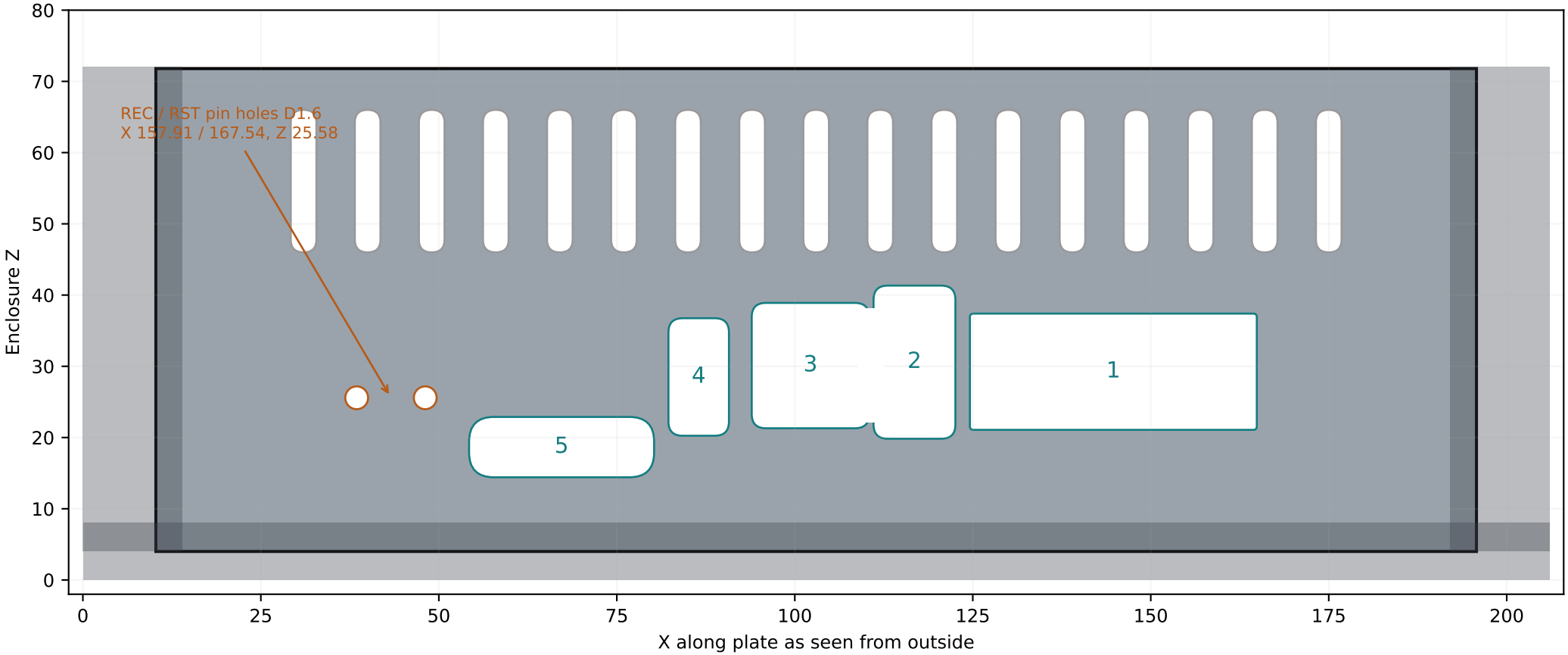
05 — Front plate WITH GMSL (viewed from outside)



1. audio: enclosure X 46.15..64.15, Z 20.08..29.08, r4.5. One obround over both jacks, centred on the jack axes; sized for two D8 plug overmolds + 0.5.
2. gmsl: enclosure X 66.20..99.80, Z 21.00..38.40, r2.0. Two candidate FM4 mini-FAKRA mouths; XY provisional (unchanged from Rev 0.4)
3. lan_1g: enclosure X 104.38..121.33, Z 20.78..36.57, r1.5. Centred on the J5 face; clears the RJ45 plug body and latch lever on insertion + 0.5; face LEDs visible.
4. lan_10g: enclosure X 124.22..141.82, Z 20.73..36.43, r1.5. Centred on the J12 face; clears the RJ45 plug body and latch lever on insertion + 0.5.
5. led: enclosure X 142.94..146.88, Z 22.20..31.15, r1.0. LED4 light pipes x 128.437..131.382, z 2.703..10.653 at y -1.5 (+0.5); bridged to neighbours when the web is < min_panel_web. Measured by slicing references/hardware/carrier-full.stl (+ open shells) at the mating face, 2026-10-25; pins and pegs below the PCB excluded.
6. dc_in: enclosure X 148.79..170.11, Z 20.93..36.93, r0.5. Sized and centred on the mated screw-flange plug, which is 15 mm tall on an 8.4 mm header and passes through the plate (body 9.9 mm outside the header face). Wire the plug before inserting it; its clamp screws face up.

3 mm plate printed exterior face down, 0.6 × 45° lead-in on every window. Plate inner face 0.37 (front) / 0.25 (rear) from the most protruding carrier part. Shaded: corner posts and sill that frame the plate. Windows narrower than a 1.8 mm web are bridged.

06 — Rear plate (viewed from outside)



1. io_terminal: enclosure X 41.11..81.41, Z 21.08..37.40, r0.5. Centred on the mated J25 plug face + 0.5.
2. hdmi: enclosure X 83.44..94.94, Z 19.83..41.33, r2.0. Centred on the J8 mouth; 11 x 21 rotated overmold + 0.25.
3. usb_a: enclosure X 95.55..112.05, Z 21.30..38.90, r2.0. Two 16 x 8 USB-A overmolds + 0.25.
4. usb_a_vertical: enclosure X 115.25..123.75, Z 20.25..36.75, r2.0. J26 corrected envelope; 8 x 16 rotated overmold + 0.25. Verify on hardware.
5. usb_c: enclosure X 125.76..151.75, Z 14.42..22.89, r3.5. Two USB-C overmolds + 0.25, side by side.

3 mm plate printed exterior face down; 0.6 × 45° lead-in on every window. Plate inner face 0.37 (front) / 0.25 (rear) from the most protruding carrier part.
Shaded: corner posts and sill that frame the plate. Windows narrower than a 1.8 mm web are bridged.

07 — Print orientation and assembly order

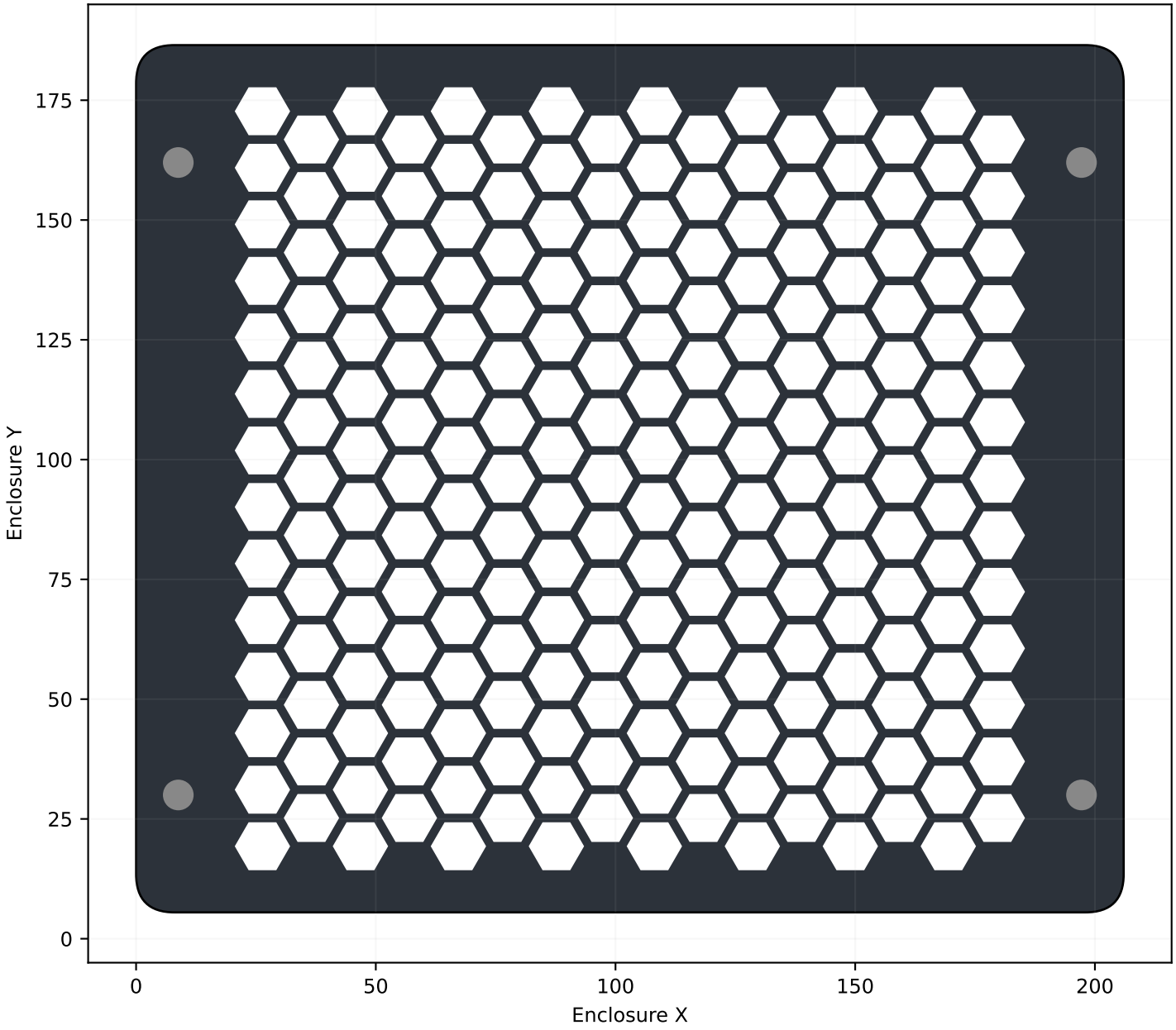
P1 / P1M: floor on bed. 206 × 181 (P1M 206 × 233) × 76.5. Columns, posts and channels rise from the floor.
P2 lid: roof on bed, 206 × 181 × 40. Hex webs grow from the bed; ledges are 45° in this orientation.
P3 / P4 / P5: exterior face on bed, 3 mm. Legends and window lead-ins print on the bed face.
P10 hatch: exterior face on bed; its V tongues have 45° undersides.
Coupons first: P11 (M3 pilot), P12/P13 (plate channel), P14 (countersink seating).

Assembly: 1. Heat-set 4 board inserts + 4 lid inserts. 2. Place PCB; 4 × M3 × 10 button heads.
3. Fit module and cooler; drive the 4 M3 × 35 flat heads up the floor pockets into the heatsink standoffs (8 threads).
4. Fit 4 RP-SMA bulkheads (WIFI 1/2 left wall, CELL 1/2 right wall, Z62), thread flat up, nut outside. Slide front, rear plates and J17 hatch down t
For wall/robot use, bolt P1M down first: the lid screws stay accessible from above.

Unresolved: filament, creep of printed columns under clamp load, insert pullout, vibration and load capacity.
Unresolved: exact cooler (thread, original screw length), GMSL XY, J26 USB-A model, user cable overmolds.

Printer/profile _____ Material _____ Rail fit _____
Countersink seat _____ Cooler screw length _____ Cable fit _____

08 — Lid: finger-safe honeycomb



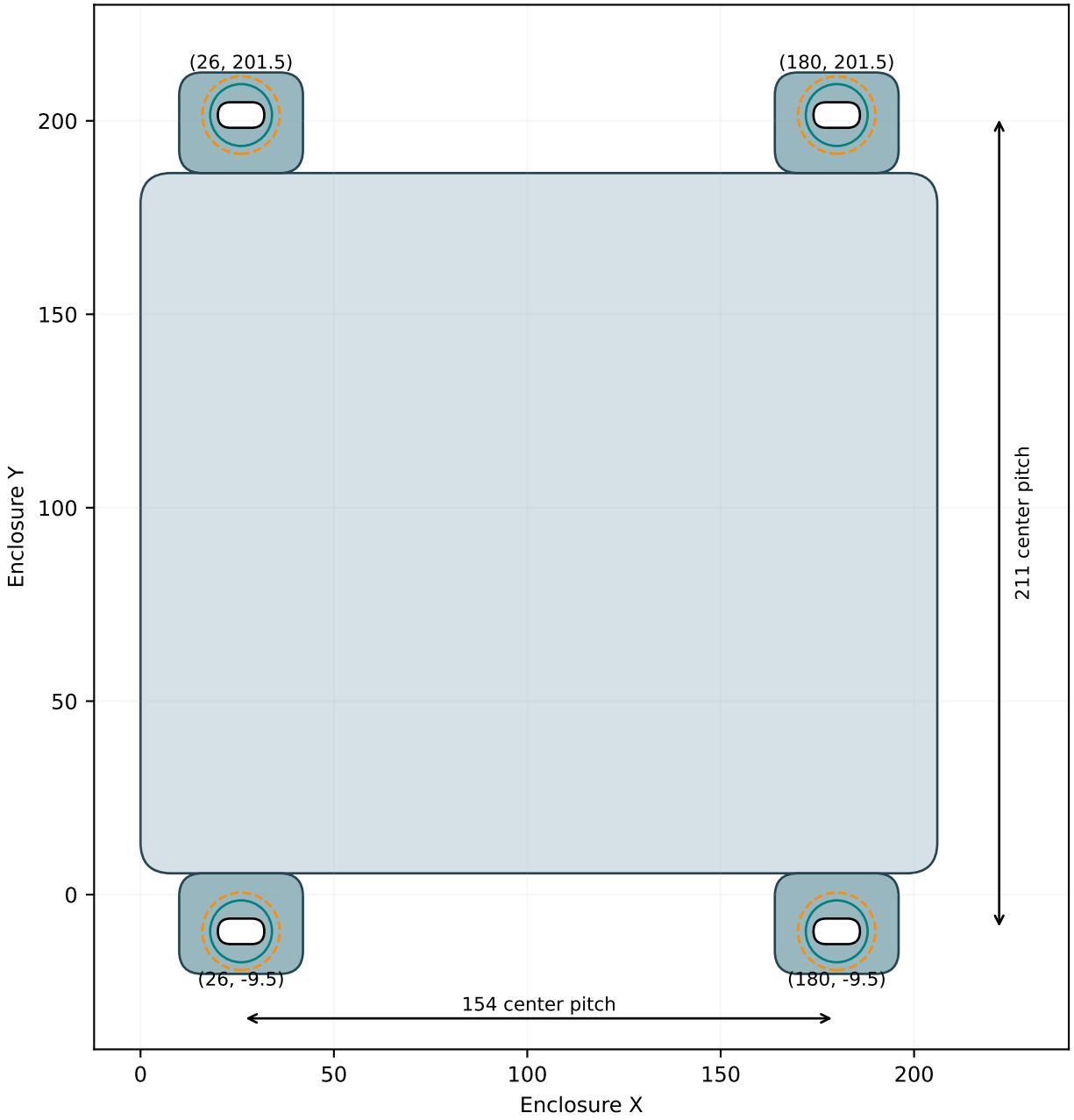
216 cells, 10 mm across flats
1.8 mm webs, 3.5 mm deep
Fan-side inlet chamfer 0.6 × 1
(11.2 A/F at the underside)
Open area 18706 mm² (50.24% of plan)
+ 60 wall slots on the lid

12 mm test finger cannot pass
(Rev 0.4: 25 × 29 mm cells)

Roof underside Z93.5
Cooler gap 1.942

Grey: M3 counterbores D6.4

09 — Mounting tabs (P1M): screw, washer and driver access

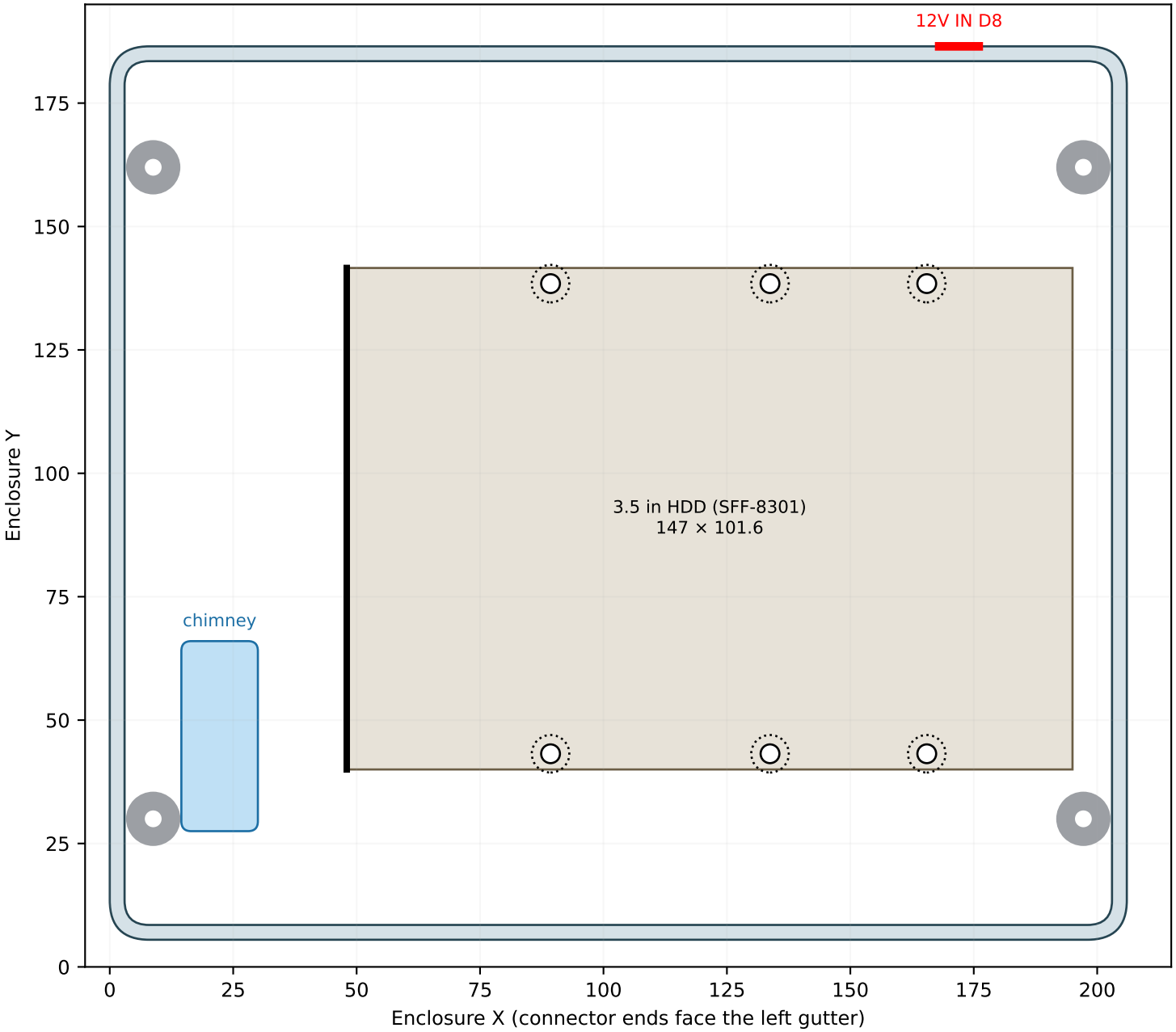


Tabs: 32 wide / 8 thick
Projection: 26; R6 plan corners
45° gusset to the frame
Slots: 12 along X / 6.6 wide

Teal: 16 mm washer envelope
Orange: 20 mm driver envelope
Nothing overhangs the tabs
(flush lid)

P1M print: 206 × 233 × 76.5
P1S Y margin 11.5 each side

10 — Storage tray 3.5 in (top view, drives from below)

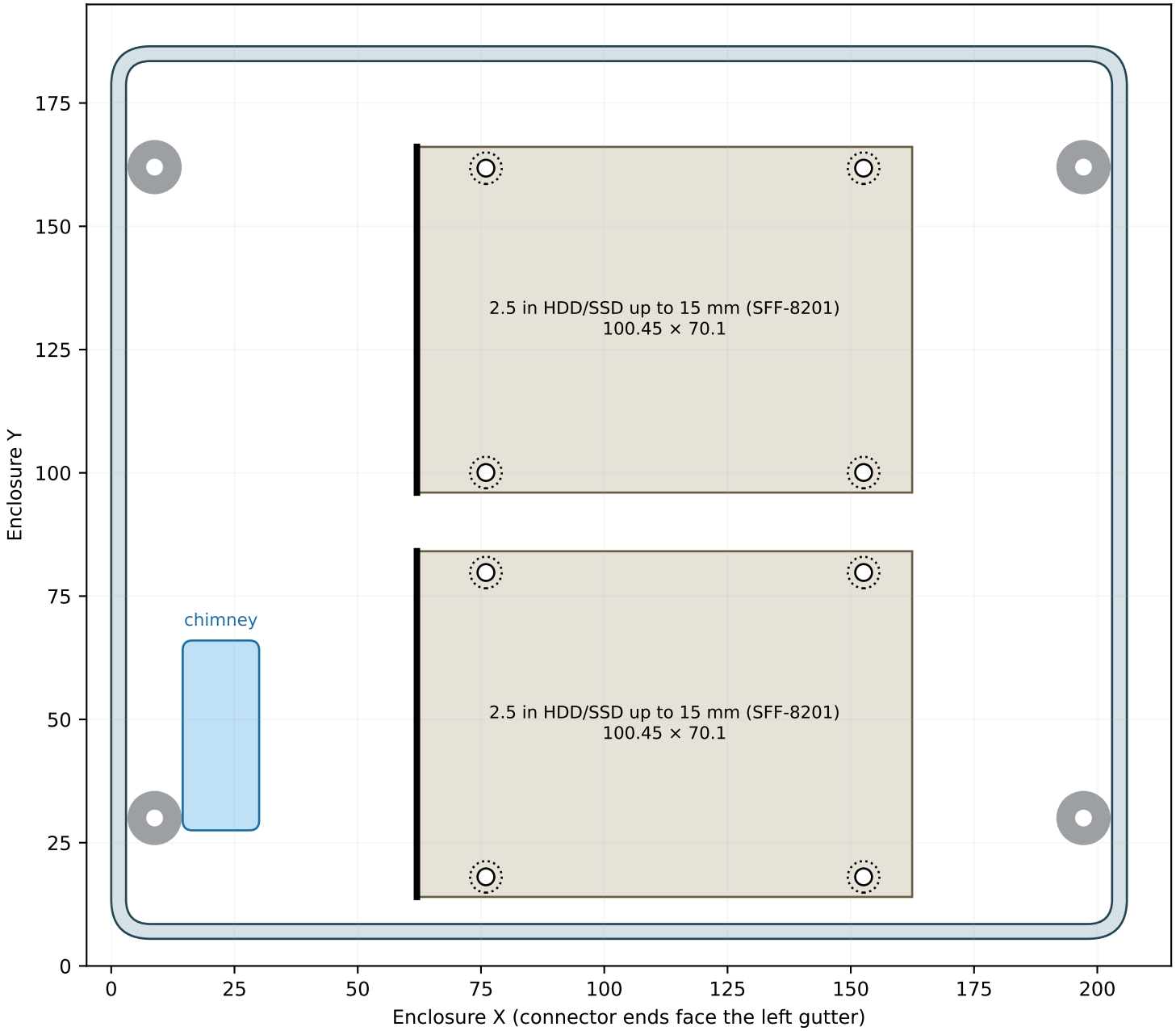


Height 32; floor 3 + pads 1.5
Standoffs: M3 M-F 30 mm
(5.5 AF, 6 mm male) in D6.6 bores,
2 mm ledge under the part above

Drive screws from below:
6-32 x 3/16 in pan head
clearance D3.8, counterbore
D7.6 × 2.6

Bay at Y40: holes 41.28, 85.73, 117.48
from the connector end (X48); lateral
3.18, 98.43.

11 — Storage tray 2.5 in (top view, drives from below)



Height 22; floor 3 + pads 1.5
Standoffs: M3 M-F 20 mm
(5.5 AF, 6 mm male) in D6.6 bores,
2 mm ledge under the part above

Drive screws from below:
M3 x 4 ISO 7380 button head (1.65 mm head fits the 2.2
clearance D3.4, counterbore
D6.4 × 2.2

Bay at Y14: holes 14, 90.6 from the
connector end (X62); lateral 4.07,
65.79.
Bay at Y96: holes 14, 90.6 from the
connector end (X62); lateral 4.07,
65.79.

12 — Under the carrier: card envelopes and SATA plug drops

envelope	ref	role	enclosure box x0, y0, z0, x1, y1, z1	pre-relief overlap mm3	gap after relief	above floor
mini_pcie_card	J13	card	16, 82, 10, 47, 139.5, 20	0	6	6
m2_b_card	J11	card	19.4, 71.5, 11, 43.5, 129, 20	0	7	7
m2_e_card	J14	card	33.5, 38, 12.5, 64.5, 62, 20	74.95	0.5	8.5
sata_plugs	J21/J24	plug_drop	14.7, 30, -10, 24, 64.5, 16.2	1161.47	-	-
sata_power_plugs	J22/J23	plug_drop	23, 36.4, -10, 28.8, 58.4, 16.2	536.06	-	-

mini_pcie_card: ST-PC-002 right-angle Mini PCIe on the underside (solid 4851, X1.53-31.43 Y67.65-76.74). Full-size 30 x 50.95 card runs toward +Y; latch/standoff bodies 5165/5166 at Y117.4-123. Depth allows a 4G/LoRa card stacked over the M.2 card plus U.FL plugs.
m2_b_card: M.2 Key B (solid 4849, X5.49-27.39 Y56.91-65.61) under the Mini PCIe card; 3042/3052 per silkscreen, card toward +Y to Y113.
m2_e_card: M.2 Key E (X48.93-58.68 Y21.75-48.27); 2230 card toward -X to the standoff body 4820 at X23.9 Y35; U.FL plugs included.
sata_plugs: Two vertical SATA 7-pin receptacles on the underside (5063/5064; datasheet photo). Cable plugs mate from below; the zone drops through the floor so cables go straight down into storage modules.
sata_power_plugs: JST PH 2.0 4-pin vertical headers (5061/5062): pin 1 NC, pins 2-3 GND, pin 4 SATA_5V (schematic sheet 27; Q59-switched VDD_5V_SYS, 2 A bead per port). No 12 V: VDD_12V feeds only the expansion connector (sheet 3).

Under-board space 16 mm; stacked cards need 10 mm. Board SATA power is 5 V only; 3.5 in drives need 12 V (tray 12V IN).